

$$\rightarrow \begin{pmatrix} 3/4 mg & 1/2 mg & -1/4 mg & 0 \\ 0 & 7/4 mg & 0 & -3/8 mg \\ 0 & 1/5 mg & 1/4 mg & -1/4 mg \\ 0 & 0 & 0 & 1/4 mg \end{pmatrix} \dots$$

Домашняя работа №2

№19.12

$$H = \frac{1}{2}(p_1^2 + p_2^2) + \frac{a}{4}(q_1 - q_2)^2 + \frac{b}{4}(q_1 + q_2)^2$$

$$H \Delta L = \dot{q} p - H | \bar{p} = \varphi(\dot{q}, \bar{p}, t)$$

$$\dot{q}_1 = \frac{\partial H}{\partial p_1} = p_1$$

$$\dot{q}_2 = + \frac{\partial H}{\partial p_2} = + p_2$$

$$L = (\dot{q}_1, \dot{q}_2) \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} - \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2) - \frac{a}{4}(q_1 - q_2)^2 - \frac{b}{4}(q_1 + q_2)^2 =$$

$$L = (\dot{q}_1, \dot{q}_2) \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \frac{a}{4}(q_2 - q_1)^2 - \frac{b}{4}(q_1 + q_2)^2 = \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2) - \frac{a}{4}(q_2 - q_1)^2 - \frac{b}{4}(q_1 + q_2)^2$$

$$\text{Ответ: } L = \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2) - \frac{a}{4}(q_1 - q_2)^2 - \frac{b}{4}(q_1 + q_2)^2$$

№19.21

Дано:

$$m_0$$

$$L = -m_0 c^2 \sqrt{1 - \frac{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}{c^2}}$$

Канонич. уравнения - ?

Закон движения - ?

Решение:

$$H = \dot{q} p - L | \dot{q} = \varphi(q, p, t)$$

$$\bar{p} = \frac{\partial L}{\partial \dot{q}}$$

$$p_x = \frac{\partial L}{\partial \dot{x}} = + m_0 c^2 \cdot \frac{\dot{x}}{c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2} =$$

$$= m_0 c \cdot \frac{\dot{x}}{\sqrt{c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2}}$$

$$\text{Аналогично: } p_y = m_0 c \cdot \frac{\dot{y}}{\sqrt{c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2}}, p_z = m_0 c \cdot \frac{\dot{z}}{\sqrt{c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2}}$$

$$\dot{x} = p_x t = m_0 c^2 \frac{\dot{x}}{c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2}$$

$$p_x^2 = m_0^2 c^2 \frac{\dot{x}^2}{c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2}, p_y^2 = m_0^2 c^2 \frac{\dot{y}^2}{c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2}, p_z^2 = m_0^2 c^2 \frac{\dot{z}^2}{c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2}$$

$$\begin{cases} m_0^2 c^2 \ddot{x}^2 = p_x^2 (c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2) \\ m_0^2 c^2 \ddot{y}^2 = p_y^2 (c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2) \\ m_0^2 c^2 \ddot{z}^2 = p_z^2 (c^2 - \dot{x}^2 - \dot{y}^2 - \dot{z}^2) \end{cases} \begin{cases} \ddot{x}^2 (m_0^2 c^2 + p_x^2) + p_x^2 \ddot{y}^2 + p_x^2 \ddot{z}^2 = p_x^2 c^2 \\ p_y^2 \ddot{x}^2 + (m_0^2 c^2 + p_y^2) \ddot{y}^2 + p_y^2 \ddot{z}^2 = p_y^2 c^2 \\ p_z^2 \ddot{x}^2 + p_z^2 \ddot{y}^2 + (m_0^2 c^2 + p_z^2) \ddot{z}^2 = p_z^2 c^2 \end{cases}$$

$$A = \begin{pmatrix} m_0^2 c^2 + p_x^2 & p_x^2 & p_x^2 \\ p_y^2 & m_0^2 c^2 + p_y^2 & p_y^2 \\ p_z^2 & p_z^2 & m_0^2 c^2 + p_z^2 \end{pmatrix}, B = \begin{pmatrix} p_x^2 c^2 \\ p_y^2 c^2 \\ p_z^2 c^2 \end{pmatrix}, A^{-1} = \begin{pmatrix} \frac{m_0^2 c^2}{p_x^2} + 1 & 1 & 1 \\ 1 & \frac{m_0^2 c^2}{p_y^2} + 1 & 1 \\ 1 & 1 & \frac{m_0^2 c^2}{p_z^2} + 1 \end{pmatrix}$$

$$\Delta = \begin{pmatrix} \frac{m_0^4 c^4}{p_x^2 p_y^2} + \frac{m_0^2 c^2}{p_x^2} + \frac{m_0^2 c^2}{p_y^2} \\ \frac{m_0^4 c^4}{p_x^2 p_z^2} + \frac{m_0^2 c^2}{p_x^2} + \frac{m_0^2 c^2}{p_z^2} \\ \frac{m_0^4 c^4}{p_y^2 p_z^2} + \frac{m_0^2 c^2}{p_y^2} + \frac{m_0^2 c^2}{p_z^2} \end{pmatrix} \begin{pmatrix} \frac{m_0^2 c^2}{p_x^2} + 1 \\ \frac{m_0^2 c^2}{p_y^2} + 1 \\ \frac{m_0^2 c^2}{p_z^2} + 1 \end{pmatrix} - \begin{pmatrix} \frac{m_0^2 c^2}{p_x^2} + 1 - 1 \\ 1 - \frac{m_0^2 c^2}{p_y^2} - 1 \\ 1 - \frac{m_0^2 c^2}{p_z^2} - 1 \end{pmatrix} = \frac{m_0^4 c^4}{p_x^2 p_y^2 p_z^2} + \frac{m_0^2 c^2}{p_x^2 p_y^2} + \frac{m_0^2 c^2}{p_x^2 p_z^2} + \frac{m_0^2 c^2}{p_y^2 p_z^2} + \frac{m_0^4 c^4}{p_x^2 p_y^2}$$

$$= \frac{m_0^4 c^4 + m_0^4 c^4 (p_y^2 + p_x^2 + p_z^2)}{p_x^2 p_y^2 p_z^2}$$

$$\Delta x = \begin{vmatrix} c^2 & 1 & 1 \\ c^2 & \frac{m_0^2 c^2}{p_y^2} + 1 & 1 \\ c^2 & 1 & \frac{m_0^2 c^2}{p_z^2} + 1 \end{vmatrix} = \left(\frac{m_0^4 c^4}{p_y^2 p_z^2} + \frac{m_0^2 c^2}{p_y^2} + \frac{m_0^2 c^2}{p_z^2} \right) c^2 - \frac{m_0^4 c^4}{p_z^2} + \frac{m_0^4 c^4}{p_y^2}$$

$$- \frac{m_0^4 c^4}{p_y^2} = \frac{m_0^4 c^4}{p_y^2 p_z^2}$$

$$\Delta y = \frac{m_0^4 c^4}{p_x^2 p_z^2}, \Delta z = \frac{m_0^4 c^4}{p_x^2 p_y^2}$$

$$\ddot{x}^2 = \frac{\Delta x}{\Delta} = \frac{\frac{m_0^4 c^4}{p_y^2 p_z^2}}{\frac{m_0^4 c^4 + m_0^4 c^4 (p_y^2 + p_x^2 + p_z^2)}{p_x^2 p_y^2 p_z^2}} = \frac{p_x^2 c^2}{m_0^2 c^2 + p_y^2 + p_x^2 + p_z^2}$$

$$\ddot{y}^2 = \frac{\Delta y}{\Delta} = \frac{\frac{m_0^4 c^4}{p_x^2 p_z^2}}{\frac{m_0^4 c^4 + m_0^4 c^4 (p_y^2 + p_x^2 + p_z^2)}{p_x^2 p_y^2 p_z^2}} = \frac{p_y^2 c^2}{m_0^2 c^2 + p_y^2 + p_x^2 + p_z^2}$$

$$H = \left(\frac{p_x c}{\sqrt{m_0^2 c^2 + p_y^2 + p_x^2 + p_z^2}} + \frac{p_y c}{\sqrt{m_0^2 c^2 + p_y^2 + p_x^2 + p_z^2}} + \frac{p_z c}{\sqrt{m_0^2 c^2 + p_y^2 + p_x^2 + p_z^2}} \right) \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} + \frac{m_0^3 c^3}{\sqrt{m_0^2 c^2 + p_y^2 + p_x^2 + p_z^2}} =$$

$$= c \cdot \sqrt{m_0^2 c^2 + p_x^2 + p_y^2 + p_z^2}$$

$$\text{Канонические уравнения: } \dot{q} = \frac{\partial H}{\partial p} \left(\dot{x} = \frac{\partial H}{\partial p_x}, \dot{y} = \frac{\partial H}{\partial p_y}, \dot{z} = \frac{\partial H}{\partial p_z} \right)$$

Выве уравнения уже написаны.

$$\dot{\vec{p}} = -\frac{\partial H}{\partial \vec{q}} = 0 \quad (\text{в } H \text{ нет } x, y, z) \Rightarrow \dot{p}_x = \dot{p}_y = \dot{p}_z = 0 \Rightarrow \text{импульсы}$$

сохраняются $\Rightarrow v = \text{const} \Rightarrow$ движение равномерное, прямолинейное

$$x(t) = x_0 + v_x \cdot t = x_0 + \dot{x} t = x_0 + \frac{p_x c}{\sqrt{m_0^2 c^2 + p_x^2 + p_y^2 + p_z^2}} \cdot t$$

$$\vec{r}(t) = \vec{r}_0 + \frac{\vec{p} c}{\sqrt{m_0^2 c^2 + p^2}} t$$

№ 19.23

Дано:

m, R

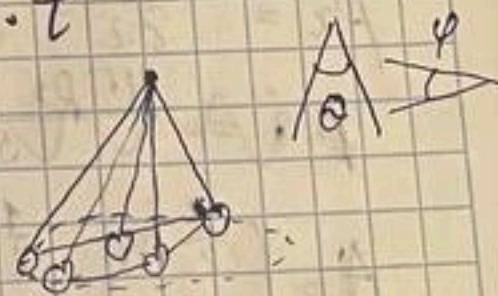
канонич.
уравнен.

Решение:

Обобщен. координаты: θ и φ

$$T = \frac{1}{2} m (\dot{R}^2 + R^2 \sin^2 \varphi \dot{\varphi}^2) \quad T = \frac{1}{2} m (R'^2 + R^2 \sin^2 \theta \dot{\theta}^2)$$

$$V = -mgR \cos \theta$$



$$\begin{cases} p_\theta = \frac{\partial L}{\partial \dot{\theta}} = m R^2 \dot{\theta} & \dot{\theta} = \frac{p_\theta}{m R^2} \\ p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = m R^2 \sin^2 \theta \dot{\varphi} & \dot{\varphi} = \frac{p_\varphi}{m R^2 \sin^2 \theta} \end{cases} \quad \text{канонич. уравнения}$$

$$\begin{cases} p_\theta = \frac{\partial L}{\partial \dot{\theta}} = m R^2 \dot{\theta} & \dot{\theta} = \frac{p_\theta}{m R^2} \\ p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = m R^2 \sin^2 \theta \dot{\varphi} & \dot{\varphi} = \frac{p_\varphi}{m R^2 \sin^2 \theta} \end{cases} \quad \text{канонич. уравнения}$$

N 19.47

Dano:

Решение:

$$P = Cx^2 + Cx^2\dot{\psi}^2$$

$$T = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m (\dot{\theta} + r \dot{\varphi})^2 = \frac{3}{4} m \dot{r}^2 + \frac{1}{4} m r^2 \dot{\varphi}^2 + \frac{1}{2} m \dot{r} \dot{\varphi} r$$

$$L = T - V = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m r^2 \dot{\phi}^2 +$$

$$+ \frac{1}{2} m \dot{\chi}^2 - c \chi^2 - c r^2 \varphi^2$$

$$\left\{ \begin{aligned} p_r &= \frac{\partial L}{\partial \dot{r}} = \frac{3}{2} m \dot{r} + \frac{1}{2} m \dot{\varphi} r \\ p_\varphi &= \frac{\partial L}{\partial \dot{\varphi}} = \frac{1}{2} m r^2 \dot{\varphi} \end{aligned} \right. \quad \dot{\varphi} = \frac{2 p_\varphi}{m r}$$

$$\begin{cases} p_r = \frac{\partial L}{\partial \dot{r}} = \frac{3}{2} m \dot{r} + \frac{1}{2} m \dot{\varphi} r \\ p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = \frac{1}{2} m r^2 \dot{\varphi} + \frac{1}{2} m \dot{r} r = \frac{r p_\varphi}{m r^2} = \frac{p_\varphi}{3 m} - \frac{p_r}{3 m} \end{cases}$$

$$H \approx (\dot{q}, p) - L = \left(\frac{2p_x^2}{3m} - \frac{2p_y^2}{3mr} \right) - \frac{2p_y^2}{mr^2} \left(\frac{p_x}{p_y} \right) - \frac{3}{4m} \left(\frac{2p_x^2}{3m} - \frac{2p_y^2}{3mr} \right)^2$$

$$= \frac{2p_x^2}{3m} - \frac{2p_y^2}{3mr} - \frac{2p_y^2}{mr^2} \cdot \frac{p_x}{p_y} - \frac{3}{4m} \left(\frac{2p_x^2}{3m} - \frac{2p_y^2}{3mr} \right)^2$$

$$= \frac{2p_x^2}{3m} - \frac{2p_y^2}{3mr} - \frac{2p_y^2}{mr^2} \cdot \frac{p_x}{p_y} - \frac{3}{4m} \left(\frac{2p_x^2}{3m} - \frac{2p_y^2}{3mr} \right)^2$$

$$= \frac{2p_x^2}{3m} - \frac{2p_y^2}{3mr} - \frac{2p_y^2}{mr^2} \cdot \frac{p_x}{p_y} - \frac{3}{4m} \left(\frac{2p_x^2}{3m} - \frac{2p_y^2}{3mr} \right)^2$$

$$\dot{\rho}_2 = \frac{\partial H}{\partial z} = 2cz, \quad \dot{\varphi} = 2\varphi c r^2$$

Handwritten graph showing a periodic wave on a grid. The wave is labeled with 4π and 5π at the first peak and trough, and 6π and 7π at the second peak and trough. The wave is drawn on a grid with a horizontal axis labeled x and a vertical axis labeled y .

$$A = \begin{pmatrix} \frac{3}{2}m & \frac{1}{2}m \\ \frac{1}{2}m^2 & \frac{1}{2}m \end{pmatrix} \quad A = \begin{pmatrix} \frac{3}{2}m & \frac{1}{2}m\tau \\ \frac{1}{2}m\tau & \frac{1}{2}m\tau^2 \end{pmatrix} \quad b = \begin{pmatrix} p_2 \\ p_4 \end{pmatrix}$$

$$\Delta = \frac{3}{4} m^2 r^2 - \frac{1}{4} m^2 r^2 = \frac{1}{2} m^2 r^2$$

$$\Delta \dot{z} = \frac{1}{2} m r^2 p_z - \frac{1}{2} m r p_\psi; \quad \Delta \dot{\psi} = \frac{3}{2} m r p_\psi - \frac{1}{2} m r p_z$$

$$\begin{cases} \dot{x} = \frac{\Delta x}{\Delta} = \frac{p_x}{m} - \frac{p_\varphi}{mr} \\ \dot{\varphi} = 3 \frac{p_\varphi}{mr^2} - \frac{p_x}{mr} \\ H = \left(\frac{p_x}{m} - \frac{p_\varphi}{mr} \right)^2 + \frac{3p_\varphi^2}{mr^2} - \frac{p_x p_\varphi}{mr} - \frac{3}{4} m \left(\frac{p_x}{m} - \frac{p_\varphi}{mr} \right)^2 - \frac{1}{4} m r^2 \\ \cdot \left(\frac{3p_\varphi}{mr^2} - \frac{p_x}{mr} \right)^2 - \frac{1}{2} m \left(\frac{p_x}{m} - \frac{p_\varphi}{mr} \right) \left(\frac{3p_\varphi}{mr^2} - \frac{p_x}{mr} \right) - c x^2 - c r^2 \varphi^2 \end{cases}$$

$$\dot{p}_x = -\frac{\partial H}{\partial x} = 2cx, \quad \dot{p}_\varphi = -\frac{\partial H}{\partial \varphi} = 2cr^2 \varphi$$

N 20.10

$$\varphi = q \cos(\omega t) + \frac{p}{\omega} \sin(\omega t)$$

$$\varphi = p \cos \omega t - q \omega \sin(\omega t)$$

$$\{ \varphi, \varphi \} = \frac{\partial \varphi}{\partial q} \cdot \frac{\partial \varphi}{\partial p} - \frac{\partial \varphi}{\partial p} \cdot \frac{\partial \varphi}{\partial q} = \cos \omega t \cdot \cos \omega t - \frac{\sin \omega t}{\omega} \cdot (-\omega \sin \omega t) = \cos^2 \omega t + \sin^2 \omega t = 1$$

$$\cdot (-\omega \sin \omega t) = \cos^2 \omega t + \sin^2 \omega t = 1$$

N 20.14

$$\begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} \end{cases} \quad \{ q_i, H \} = \frac{\partial q_i}{\partial q_j} \frac{\partial H}{\partial p_j} - \frac{\partial q_i}{\partial p_j} \frac{\partial H}{\partial q_j} = 1 \cdot \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial q_i} \frac{\partial q_j}{\partial p_j} \Rightarrow$$

$$\{ p_i, H \} = \frac{\partial p_i}{\partial q_j} \frac{\partial H}{\partial p_j} - \frac{\partial p_i}{\partial p_j} \frac{\partial H}{\partial q_j} = -\frac{\partial H}{\partial q_i}$$

$$\begin{cases} \dot{q}_i = \{ q_i, H \} \\ \dot{p}_i = \{ p_i, H \} \end{cases} \quad \text{— канонич. уравнен.} \quad \text{через скобки Пуассона}$$

N 20.24

$$H = H[\varphi_1(q_1, p_1), \dots, \varphi_n(q_n, p_n), t]$$

$$\varphi_k(q_k, p_k) = d_k \quad \text{— ПУ — показать}$$

$$\begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} = \frac{\partial \varphi_1}{\partial p_i} \frac{d\varphi_1}{d\varphi_1} + \dots + \frac{\partial \varphi_n}{\partial p_i} \frac{d\varphi_n}{d\varphi_i} \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} = -\frac{\partial \varphi_1}{\partial q_i} \frac{d\varphi_1}{d\varphi_i} - \dots - \frac{\partial \varphi_n}{\partial q_i} \frac{d\varphi_n}{d\varphi_i} \end{cases}$$

$$\begin{cases} \dot{q}_i = \frac{\partial \varphi_1}{\partial p_i} = \frac{\partial \varphi_1}{\partial p_i} \cdot \frac{\partial \varphi_1}{\partial q_i} \\ \dot{p}_i = -\frac{\partial \varphi_1}{\partial q_i} = -\frac{\partial \varphi_1}{\partial q_i} \cdot \frac{\partial \varphi_1}{\partial p_i} \end{cases} \quad \begin{cases} \dot{q}_i \cdot \frac{\partial \varphi_1}{\partial q_i} = \frac{\partial \varphi_1}{\partial p_i} \frac{\partial \varphi_1}{\partial q_i} \\ \dot{p}_i \cdot \frac{\partial \varphi_1}{\partial p_i} = -\frac{\partial \varphi_1}{\partial q_i} \frac{\partial \varphi_1}{\partial p_i} \end{cases} +$$

$$\frac{\partial \varphi_1}{\partial q_i} \cdot \frac{\partial q_i}{\partial t} + \frac{\partial \varphi_1}{\partial p_i} \cdot \frac{\partial p_i}{\partial t} = 0 \quad \frac{d\varphi_1(p_i, q_i)}{dt} = 0 \quad \varphi_1 = \text{const}$$

Аналогично для остальных $\Rightarrow \varphi_i(q_i, p_i) = d_i = \text{const}$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} = \frac{\partial H}{\partial \dot{q}_i} \cdot \frac{\partial \dot{q}_i}{\partial p_i} = 1 \cdot \frac{\partial \dot{q}_i}{\partial p_i} = \frac{\partial \dot{q}_i}{\partial p_i}$$

$$\text{или } q_i(t) = \int \frac{\partial \dot{q}_i}{\partial p_i} dt - \text{уравн. движения.}$$

N 20.36

Дано:

$$m_1, \dots, m_n$$

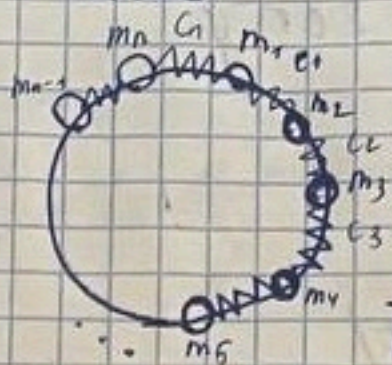
$$q_i = r \varphi_i = ?$$

$$\tilde{q}_i = q_i(q, d)?$$

L не мен

$$\sum_{i=1}^n m_i q_i = \Pi U$$

Решение:



$$\Pi = \frac{C_1 q_1^2}{2} + \dots + \frac{C_{n-1} q_{n-1}^2}{2} + \frac{C_n q_n^2}{2}$$

$$\frac{\partial \Pi}{\partial q_i} = 2 C_i q_i = 0 \quad q_i = 0 = r \varphi_i$$

$$T = \frac{m_1 \dot{q}_1^2}{2} + \dots + \frac{m_n \dot{q}_n^2}{2}$$

$$L = T - \Pi = \frac{m_1 r^2 \dot{\varphi}_1^2}{2} + \dots + \frac{m_n r^2 \dot{\varphi}_n^2}{2} - \frac{C_1 r^2 \varphi_1^2}{2} - \dots - \frac{C_n r^2 \varphi_n^2}{2}$$

$$\tilde{q}_i = q_i + d = \text{const}$$

$$\Pi = \frac{1}{2} C_n (q_1 - q_n)^2 + \frac{1}{2} (q_2 - q_1)^2 C_1 + \dots + \frac{1}{2} C_{n-1} (q_n - q_{n-1})^2$$

$$L = T - \Pi = \frac{1}{2} m_1 r^2 \dot{\varphi}_1^2 + \dots + \frac{1}{2} m_n r^2 \dot{\varphi}_n^2 - \frac{1}{2} (q_2 - q_1)^2 C_1 - \dots - \frac{1}{2} C_n (q_1 - q_n)^2$$

$$\forall i \quad \tilde{q}_i = q_i + d, \quad d = \text{const}$$

$$\dot{\tilde{q}}_i = \dot{q}_i, \quad q_{i+1} - q_i = q_{i+1} + d - q_i - d = q_{i+1} - q_i \Rightarrow$$

$$\Rightarrow \tilde{L} = L$$

Теорема Эмми-Кепер:

$$\text{Если } \tilde{L} dt = L dt, \text{ то } \exists \Pi U \quad \Phi = \sum p_i \frac{\partial \tilde{q}_i}{\partial d} - H \frac{\partial \tilde{t}}{\partial d} \Big|_{d=0}$$

$$\frac{\partial \tilde{E}}{\partial d} = 0, \quad \frac{\partial \tilde{q}_i}{\partial d} = 1 \Rightarrow \Phi = \sum_{i=1}^n p_i = \sum_{i=1}^n \frac{\partial L}{\partial \dot{q}_i} = \sum_{i=1}^n \left(\frac{1}{2} m_i r^2 \dot{\varphi}_i^2 \right)' =$$

$$= \sum_{i=1}^n \left(\frac{1}{2} m_i \dot{q}_i^2 \right)' = \sum_{i=1}^n m_i \dot{q}_i = \Pi U = \text{const} \quad \text{и т.д.}$$

N 21.10

Решение:

$$\partial W(q_{\text{равн}}) = 0 \Leftrightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = 0, \text{ т.к. не завис. от времени}$$

$$p = \frac{\partial L}{\partial \dot{q}} \Rightarrow \frac{d}{dt} p = 0 \Rightarrow \dot{p}_x = \dot{p}_y = \dot{p}_z = 0 \Rightarrow$$

$\Rightarrow$ прямой путь соответ. равномерного прямолинейного движения

$$r(t) = r(0) + \dot{r} t, \text{ - прямая в пространстве}$$

Дано:

$$L = -m c^2 \sqrt{1 - \frac{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}{c^2}}$$

Через любые 2

точки (x_0, y_0, z_0, t_0) ,

(x_1, y_1, z_1, t_1) , $t_0 \neq t_1$

проходим прямую

пути и только

одну

$\Rightarrow$ а для прямой: через любые 2 точки можно провести прямую и только одну ч.м.г

N21.23

Дано:

$$L = \frac{1}{2} \sum_{i=1}^n (a_i^2 \dot{q}_i^2 + b_i^2 \dot{q}_i^2)$$

$$a_i = \text{const}$$

$$b_i = \text{const}$$

$W(q_{np})$ - миним.

Решение:

$$W(q + \delta q) = W(q + \delta q) = \int L(q + \delta q, \dot{q} + \delta \dot{q}, t) dt =$$

$$= \frac{1}{2} \int \sum_{i=1}^n (a_i^2 (\dot{q}_i + \delta \dot{q}_i)^2 + b_i^2 (\dot{q}_i + \delta \dot{q}_i)^2) dt = \frac{1}{2} \int \sum_{i=1}^n (a_i^2 \dot{q}_i^2 +$$

$$+ b_i^2 \dot{q}_i^2 + 2a_i^2 \dot{q}_i \delta \dot{q}_i + 2b_i^2 \dot{q}_i \delta \dot{q}_i) dt = \frac{1}{2} \int \sum_{i=1}^n (a_i^2 \dot{q}_i^2 + b_i^2 \dot{q}_i^2) dt +$$

$$+ (2a_i^2 \dot{q}_i \delta \dot{q}_i + 2b_i^2 \dot{q}_i \delta \dot{q}_i) dt + \frac{1}{2} \int \sum_{i=1}^n (a_i^2 \delta \dot{q}_i^2 + b_i^2 \delta \dot{q}_i^2) dt =$$

$$= \frac{1}{2} \int \sum_{i=1}^n W(q_{np}) dt + \frac{1}{2} \int \sum_{i=1}^n \delta W(q_{np}) dt + \delta W \frac{1}{2} \int \sum_{i=1}^n \delta^2 W$$

$$\delta^2 W = a_i^2 \delta \dot{q}_i^2 + b_i^2 \delta \dot{q}_i^2 \geq 0 \Rightarrow \text{так } W(q_{np}) - \min (\text{государств}$$

минимум действия) ч.м.г.

N22.6

$$I = \oint \sum_{i=1}^n [(\alpha_i p_i + \varphi_i(q_i)) \delta q_i + (\beta_i q_i + \psi_i(p_i)) \delta p_i]$$

I - инвариант. Тогда согласно пр.м. и Ха-Джунс

$$I = \oint I \bar{x}, I \bar{x} = \oint \sum_{i=1}^n p_i \delta q_i$$

$$I - \oint I \bar{x} = 0 \quad \oint \sum_{i=1}^n [(\alpha_i - \beta_i) p_i + \varphi_i(q_i)] \delta q_i + (\beta_i q_i + \psi_i(p_i)) \delta p_i = 0$$

$$\Rightarrow \exists F: \quad \frac{\partial F}{\partial p_i} = \alpha_i - \beta_i, \quad \frac{\partial F}{\partial q_i} = \beta_i q_i + \psi_i(p_i)$$

$$\beta_i q_i + \psi_i(p_i) = \frac{\partial F}{\partial p_i}$$

$$\frac{\partial F}{\partial p_i} = \frac{\partial F}{\partial q_i} \frac{\partial q_i}{\partial p_i}$$

$$\beta_i = \alpha_i - \beta_i, \quad \beta_i = \frac{\alpha_i - \beta_i}{\frac{\partial q_i}{\partial p_i}} \neq \text{const} \Rightarrow$$

(м.к. $\beta_i = -p_i + \alpha_i, \beta_i = p_i + \alpha_i$
(в общем случае нет))

$\Rightarrow$ м. не выполнят $\Rightarrow I$ не инвариант.

N22.20

$H = \varphi_3(\varphi_2[\varphi_1(q_1, p_1), q_2, p_2], q_3, p_3)$ - матрица $\Rightarrow$

$\varphi_1, \varphi_2, \varphi_3$ - ПИ - равны константам

Раз система гамильтонова, то фазовый объем

системы сохраняется, I_1, I_2, I_3 равны произвед.

фазов. объемов на $\varphi_1, \varphi_2, \varphi_3 \Rightarrow I_1, I_2, I_3$ - константы $\Rightarrow$ интегр. инвариант.

N 22.30

$$\dot{x}_1 = a_{11} x_1 + a_{12} x_2, \quad \dot{x}_2 = a_{21} x_1 + a_{22} x_2$$

Система гамильтонова, если $\dot{x}_1 = \frac{\partial H}{\partial x_2}, \quad \dot{x}_2 = -\frac{\partial H}{\partial x_1}$, т.е.

$$\frac{\partial H}{\partial x_2} \frac{\partial H}{\partial x_1} = \frac{\partial H}{\partial x_1} \frac{\partial H}{\partial x_2}; \quad \frac{\partial H}{\partial x_1} = a_{11}, \quad \frac{\partial H}{\partial x_2} = -a_{22} \Rightarrow$$

$\Rightarrow$ система гамильтонова, если $a_{11} = -a_{22}, \quad a_{11} + a_{22} = 0$

Теорема Лувилля: фаз. объем системы сохраняется
 $\Leftrightarrow \operatorname{div} \vec{F} = 0: \quad \frac{\partial f_1}{\partial x_1} + \frac{\partial f_2}{\partial x_2} = 0 \quad a_{11} + a_{22} = 0$, т.е. фаз. объем

сохран. если только система гамильтонова.

$$H = \int (a_{11} x_1 + a_{12} x_2) dx_2 = a_{11} x_1 x_2 + a_{12} \frac{x_2^2}{2}$$

$$H = -\int (a_{21} x_1 + a_{22} x_2) dx_1 = -a_{21} x_1 x_2 - a_{22} \frac{x_1^2}{2}$$

$$2H = a_{12} \frac{x_2^2}{2} - a_{21} \frac{x_1^2}{2} + 2a_{11} x_1 x_2; \quad x_1 = q, \quad x_2 = p$$

$$H = a_{12} \frac{p^2}{4} + a_{11} q p - a_{21} \frac{q^2}{4}$$

N 23.7

$$\begin{cases} \tilde{q} = -q \operatorname{ctg} p \\ \tilde{p} = 2 \ln \cos p \end{cases} \quad \begin{cases} \frac{\partial \tilde{q}}{\partial \tilde{q}} \tilde{p} - \tilde{q} = -\frac{\partial F}{\partial \tilde{q}} \\ \frac{\partial \tilde{q}}{\partial \tilde{p}} \tilde{p} = -\frac{\partial F}{\partial \tilde{p}} \end{cases}$$

$$\begin{cases} -\operatorname{ctg} p \cdot 2 \ln \cos p - \tilde{q} = -\frac{\partial F}{\partial \tilde{q}} \\ + q \cdot \frac{2 \sin p}{\sin^2 p} \cdot 2 \ln \cos p = -\frac{\partial F}{\partial \tilde{p}} \end{cases}$$

$$-\frac{\partial^2 F}{\partial \tilde{q} \partial \tilde{p}} = \frac{2 \ln \cos p}{\sin^2 p} + \frac{2 \sin p}{\cos p} \cdot \operatorname{ctg} p - c = \frac{2 \ln \cos p}{\sin^2 p} = -\frac{\partial^2 F}{\partial \tilde{p} \partial \tilde{q}}$$

$$\frac{2 \ln \cos p}{\sin^2 p} + 2 - c = \frac{2 \ln \cos p}{\sin^2 p}, \quad c = 2$$

$$F = \int (2 \operatorname{ctg} p \ln \cos p + 2p) dq = 2 \operatorname{ctg} p \ln \cos p q + 2pq$$

N 23.20

$$\begin{cases} x_1 = \rho \cos \varphi \\ x_2 = \rho \sin \varphi \\ x_3 = z \end{cases} \quad \begin{cases} \tilde{p}_i = \sum_{j=1}^n p_j \frac{\partial q_j}{\partial \tilde{q}_i} \\ q_1 = \tilde{q}_1 \cos \tilde{q}_2 \\ q_2 = \tilde{q}_1 \sin \tilde{q}_2 \\ q_3 = \tilde{q}_3 \end{cases}, \quad C=1, F=0 \quad \tilde{q} = (\rho, \varphi, z)$$

$$P_1 = \sum_{i=1}^3 P_{1i} = P_1$$

$$P_2 = P_{21} \frac{\partial z_1}{\partial \varphi} + P_{22} \frac{\partial z_2}{\partial \varphi} + P_{23} \frac{\partial z_3}{\partial \varphi}$$

$$P_2 = P_{21} \frac{\partial z_1}{\partial \varphi} + P_{22} \frac{\partial z_2}{\partial \varphi} + P_{23} \frac{\partial z_3}{\partial \varphi} = P_{21} \cos \varphi + P_{22} \sin \varphi =$$

$$= \frac{P_{21} z_1 + P_{22} z_2}{\sqrt{z_1^2 + z_2^2}} = \frac{P_{21} z_1 + P_{22} z_2}{\rho}$$

$$P_3 = P_{21} \frac{\partial z_1}{\partial \varphi} + P_{22} \frac{\partial z_2}{\partial \varphi} + P_{23} \frac{\partial z_3}{\partial \varphi} = -P_{21} \rho \sin \varphi + P_{22} \rho \cos \varphi =$$

$$= P_{22} z_1 - P_{21} z_2$$

$$J^{-1} = \frac{\rho}{z} \begin{pmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi / \rho & \cos \varphi / \rho & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$P_z = P_{23} \frac{\partial z}{\partial \varphi}$$

$$P_{21} = P_2 \frac{\partial z_1}{\partial \varphi} + P_3 \frac{\partial z_2}{\partial \varphi} + P_z \frac{\partial z_3}{\partial \varphi} =$$

$$= P_2 \cos \varphi + P_3 \sin \varphi / \rho, P_{22} = P_2 \sin \varphi + P_3 \frac{\cos \varphi}{\rho}$$

$$P_{23} = P_z$$

N 23.29

$$\tilde{q}_i = \tilde{q}_i(q, p, t), \tilde{p}_i = \tilde{p}_i(q, p, t) - \text{канон. преобр.}$$

$$q_i^* = f_1(t) \tilde{q}_i(q, p, t), p_i^* = f_2(t) \tilde{p}_i(q, p, t)$$

$$\begin{cases} \frac{\partial H^*}{\partial q_i^*} \frac{\partial q_i^*}{\partial q_i} \tilde{p}_i^* - \tilde{c}_i^* = -\frac{\partial F}{\partial q_i}, & \frac{\partial q_i^*}{\partial q_i} = f_1(t) \frac{\partial \tilde{q}_i}{\partial q_i}, p_i^* = f_2(t) \tilde{p}_i \\ \frac{\partial q_i^*}{\partial p_i} \tilde{p}_i^* = -\frac{\partial F}{\partial p_i}, & \frac{\partial q_i^*}{\partial p_i} = f_1(t) \frac{\partial \tilde{q}_i}{\partial p_i}, p_i^* = f_2(t) \tilde{p}_i \end{cases}$$

$$\begin{cases} f_1(t) f_2(t) \frac{\partial \tilde{q}_i}{\partial q_i} \tilde{p}_i - c_i = -\frac{\partial F}{\partial q_i} \\ f_1(t) f_2(t) \frac{\partial \tilde{q}_i}{\partial p_i} \tilde{p}_i = -\frac{\partial F}{\partial p_i} \end{cases}$$

$$-\frac{\partial F}{\partial q_i \partial p_i} = f_1(t) f_2(t) \left(\frac{\partial \tilde{q}_i}{\partial q_i} \frac{\partial \tilde{p}_i}{\partial p_i} + \frac{\partial \tilde{q}_i}{\partial p_i} \frac{\partial \tilde{p}_i}{\partial q_i} \right) - c =$$

$$= f_1(t) f_2(t) \left(\frac{\partial \tilde{q}_i}{\partial p_i} \frac{\partial \tilde{p}_i}{\partial q_i} + \frac{\partial \tilde{q}_i}{\partial q_i} \frac{\partial \tilde{p}_i}{\partial p_i} \right) = -\frac{\partial F}{\partial p_i \partial q_i}$$

$$c = f_1(t) f_2(t) \left(\frac{\partial \tilde{q}_i}{\partial p_i} \frac{\partial \tilde{p}_i}{\partial p_i} - \frac{\partial \tilde{q}_i}{\partial q_i} \frac{\partial \tilde{p}_i}{\partial q_i} \right). \text{ Из условия } \tilde{q}_i, \tilde{p}_i -$$

канон. преобр. получаем, что выражение в скобках =

$$= \text{const} \Rightarrow c = f_1(t) f_2(t) \cdot L. \text{ Чтобы } c = \text{const}, \text{ нужно}$$

$$f_1(t) f_2(t) = \text{const}$$

$$\text{Ответ: } f_1(t) f_2(t) = \text{const}$$

N 23.48

$$H = \frac{1}{2} \sum_{i=1}^n (p_i^2 + \omega_i^2 q_i^2), \quad \dot{p}_i = -\frac{\partial H}{\partial q_i} = -\omega_i^2 q_i$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} = p_i \Rightarrow \ddot{q}_i = \dot{p}_i \Rightarrow \ddot{q}_i = -\omega_i^2 q_i, \quad \ddot{q}_i + \omega_i^2 q_i = 0 - \text{колебания}$$

$$\begin{cases} q_i = q_i(q_i, p_i, t) \\ p_i = p_i(q_i, p_i, t) \\ q_i = p_i \\ p_i = -\omega_i^2 q_i \end{cases}$$

$$q_i = x_i + p_i + m_i$$

newton currenty gapogrub

$$A = \begin{pmatrix} 0 & 1 \\ -\omega_i^2 & 0 \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\lambda^2 + \omega_i^2 = 0 \Leftrightarrow \lambda = \pm i\omega_i$$

Содѣм. бермора:

$$\begin{pmatrix} -i\omega_i & 1 \\ -\omega_i^2 & -i\omega_i \end{pmatrix} \quad \begin{pmatrix} -i\omega_i & 1 \\ -\omega_i^2 & -i\omega_i \end{pmatrix} \quad h_1 = \begin{pmatrix} 1 \\ i\omega_i \end{pmatrix}, \quad h_2 = \begin{pmatrix} 1 \\ -i\omega_i \end{pmatrix}$$

$$OKP: \quad C_1 e^{i\omega_i t} \begin{pmatrix} 1 \\ i\omega_i \end{pmatrix} + C_2 e^{-i\omega_i t} \begin{pmatrix} 1 \\ -i\omega_i \end{pmatrix} = r = c (\cos \omega_i t + i \sin \omega_i t).$$

$$\begin{aligned} ODP: \quad C_1 \begin{pmatrix} 1 \\ i\omega_i \end{pmatrix} + C_2 e (\cos \omega_i t - i \sin \omega_i t) \begin{pmatrix} 1 \\ -i\omega_i \end{pmatrix} = \\ = (\cos \omega_i t + i \sin \omega_i t) \begin{pmatrix} 1 \\ i\omega_i \end{pmatrix} + C_2 (\cos \omega_i t - i \sin \omega_i t) \begin{pmatrix} 1 \\ -i\omega_i \end{pmatrix} = \\ = C_1 (\cos \omega_i t + i \sin \omega_i t) \begin{pmatrix} 1 \\ i\omega_i \end{pmatrix} = C_1 \begin{pmatrix} \cos \omega_i t \\ -\omega_i \sin \omega_i t \end{pmatrix} + C_2 \begin{pmatrix} \sin \omega_i t \\ \omega_i \cos \omega_i t \end{pmatrix} \end{aligned}$$

$$ODP: \quad \begin{pmatrix} q_i \\ p_i \end{pmatrix} = C_1 \begin{pmatrix} \cos \omega_i t \\ -\omega_i \sin \omega_i t \end{pmatrix} + C_2 \begin{pmatrix} \sin \omega_i t \\ \omega_i \cos \omega_i t \end{pmatrix}$$

$$C_1 = q_i(0) = q_i^0, \quad C_2 = \frac{p_i^0}{\omega_i} \Rightarrow$$

$$\begin{pmatrix} q_i \\ p_i \end{pmatrix} = \begin{pmatrix} \cos \omega_i t \\ -\omega_i \sin \omega_i t \end{pmatrix} q_i^0 + \begin{pmatrix} \sin \omega_i t \\ \omega_i \cos \omega_i t \end{pmatrix} \frac{p_i^0}{\omega_i}$$

Примерный расчет:

$$p_i \delta q_i - c p_i^0 \delta q_i^0 = -\delta F_i \quad (-\omega_i q_i^0 \sin \omega_i t + p_i^0 \cos \omega_i t) (\cos \omega_i t \delta q_i^0 + \sin \omega_i t \frac{\delta p_i^0}{\omega_i}) - c p_i^0 \delta q_i^0 = -\delta F$$

$$-\omega_i q_i^0 \delta q_i^0 \sin \omega_i t \cos \omega_i t + p_i^0 \delta q_i^0 \cos^2 \omega_i t - \omega_i q_i^0 \delta p_i^0 \sin^2 \omega_i t + \frac{p_i^0}{\omega_i} \delta p_i^0 \sin \omega_i t \cos \omega_i t - c p_i^0 \delta q_i^0 = -\delta F$$

$$\sin \omega_i t \cos \omega_i t - c p_i^0 \delta q_i^0 = -\delta F$$

Примерный расчет: $q_i \delta q_i$ и $p_i \delta p_i$ нулево:

$$\cos^2 \omega_i t - c = -\sin^2 \omega_i t \Rightarrow c = 1 \Rightarrow \text{нулевая}$$

Примерный расчет: $\frac{\partial P}{\partial p_i^0} = \frac{\partial Q}{\partial q_i^0}$ (гидрогравитация, гравитация, измерение, измерение)

$$\cos^2 \omega_i t - c = -\sin^2 \omega_i t \Rightarrow c = 1 \Rightarrow \text{нулевая}$$

$$-\frac{1}{2} \omega_i q_i^0 \sin^2 \omega_i t + p_i^0 (\cos^2 \omega_i t - 1) q_i^0 + g(p_i^0) =$$

$$= -q_i^0 p_i^0 \sin^2 \omega_i t + \frac{1}{2} p_i^0 \sin^2 \omega_i t \cos \omega_i t \cdot \frac{1}{\omega_i} + f(q_i^0) \Rightarrow$$

$$F = \frac{1}{2\omega} p_i^2 \sin \omega t \cos \omega t - \frac{1}{2} \omega q_i^2 \sin \omega t \cos \omega t - q_i p_i \sin^2 \omega t$$

Но $H_i^0 = H_i = c H_i^0 + \frac{\partial F}{\partial t} + p_i \frac{\partial q_i}{\partial t}$ при нормировке
 нормировке $H_i^0 = 0$.

N 23.71

$$\begin{cases} \tilde{q}_i = q_i^{\alpha_i} p_i^{\beta_i} \\ \tilde{p}_i = q_i^{\delta_i} p_i^{\gamma_i} \end{cases} - \text{канонич. } \alpha_i, \beta_i, \gamma_i, \delta_i = ?$$

$$\begin{cases} \frac{\partial \tilde{q}_i}{\partial q_i} \tilde{p}_i - c p_i = - \frac{\partial F}{\partial q_i} \\ \frac{\partial \tilde{q}_i}{\partial p_i} \tilde{p}_i = - \frac{\partial F}{\partial p_i} \end{cases} \begin{cases} \alpha_i q_i^{\alpha_i-1} p_i^{\beta_i} \tilde{p}_i - c p_i = - \frac{\partial F}{\partial q_i} \\ \beta_i p_i^{\beta_i-1} q_i^{\alpha_i} \tilde{p}_i = - \frac{\partial F}{\partial p_i} \end{cases}$$

$$- \frac{\partial F}{\partial q_i \partial p_i} = - \frac{\partial F}{\partial p_i \partial q_i} \quad d \ln q_i^{\alpha_i-1} p_i^{\beta_i-1} \tilde{p}_i = p_i^{\beta_i} q_i^{\alpha_i} p_i^{\beta_i-1} q_i^{\alpha_i-1}$$

$$d \ln \tilde{p}_i = d \ln p_i^{\beta_i} q_i^{\alpha_i} p_i^{\beta_i-1} q_i^{\alpha_i-1} = (\beta_i + \alpha_i) d \ln q_i + (\beta_i - \alpha_i) d \ln p_i$$

$$(\beta_i + \alpha_i) \alpha_i q_i^{\alpha_i-1} p_i^{\beta_i-1} \tilde{p}_i - c p_i = (\beta_i - \alpha_i) \beta_i p_i^{\beta_i-1} q_i^{\alpha_i}$$

$$\alpha_i + \beta_i = 1, \quad \beta_i + \alpha_i = 1 \quad (\beta_i + \alpha_i) \alpha_i - c = (\beta_i - \alpha_i) \beta_i \Rightarrow c =$$

$$= \beta_i \alpha_i - \alpha_i \beta_i = (1 - \beta_i) \alpha_i - (1 - \alpha_i) \beta_i = \alpha_i - \beta_i = \beta_i - \alpha_i$$

N 23.97

$$\tilde{q}_i = \sqrt[2\alpha_i]{p_i q_i^{1-\alpha_i}}, \quad \tilde{p}_i = - \sqrt[2\alpha_i]{(p_i)^{2\alpha_i-1} q_i^{2\alpha_i-1}}$$

$$\begin{cases} p_i = p_i(q_i, \tilde{q}_i, t) \\ \tilde{p}_i = \tilde{p}_i(q_i, \tilde{q}_i, t) \end{cases} \begin{cases} \tilde{p}_i = - \frac{\partial S}{\partial \tilde{q}_i} \\ c p_i = \frac{\partial S}{\partial q_i} \end{cases} \Rightarrow \frac{\partial S}{\partial q_i} p_i - \frac{\partial^2 S}{\partial \tilde{q}_i \partial q_i} = \frac{\partial^2 S}{\partial q_i \partial \tilde{q}_i}$$

$$q_i^{\alpha_i} = \sqrt[2\alpha_i]{p_i q_i^{1-\alpha_i}} \Rightarrow p_i = q_i^{2\alpha_i-1} \Rightarrow \tilde{p}_i = - \sqrt[2\alpha_i]{q_i^{2\alpha_i-1}}$$

$$\frac{\partial \tilde{p}_i}{\partial q_i} = - \frac{2\alpha_i-1}{2\alpha_i} q_i^{2\alpha_i-2} = - \frac{1-2\alpha_i}{2\alpha_i} \sqrt[2\alpha_i]{q_i^{2\alpha_i-1}}$$

$$c \frac{\partial p_i}{\partial q_i} = c (2\alpha_i-1) q_i^{2\alpha_i-2}$$

$$\frac{1}{2\alpha_i} \sqrt[2\alpha_i]{(p_i)^{2\alpha_i-1} q_i^{2\alpha_i-1}} = \frac{c}{\sqrt[2\alpha_i]{q_i^{2\alpha_i-1}}} \Rightarrow c \text{ const} \Rightarrow \text{не канонич.}$$

$$\tilde{q}_i^{\alpha_i} = \sqrt[2\alpha_i]{p_i q_i^{1-\alpha_i}} \Rightarrow p_i = \frac{\tilde{q}_i^{\alpha_i}}{q_i^{2\alpha_i-1}}$$

$$\tilde{p}_i = - \sqrt[2\alpha_i]{\frac{\tilde{q}_i^{\alpha_i}}{q_i^{2\alpha_i-1}}} = - \sqrt[2\alpha_i]{\frac{\tilde{q}_i^{\alpha_i}}{q_i^{2\alpha_i-1}}} = - \frac{\tilde{q}_i^{\alpha_i-1}}{q_i^{\alpha_i-2}}$$

$$\frac{\partial \tilde{p}_i}{\partial q_i} = - \alpha_i \frac{\tilde{q}_i^{\alpha_i-1}}{q_i^{\alpha_i-1}}, \quad c \frac{\partial p_i}{\partial q_i} = \frac{c \alpha_i}{\sqrt[2\alpha_i]{q_i^{2\alpha_i-1}}} \Rightarrow \frac{c}{\sqrt[2\alpha_i]{q_i^{2\alpha_i-1}}} = 1 \Rightarrow c = \sqrt[2\alpha_i]{q_i^{2\alpha_i-1}}$$

$$p_i = \frac{\tilde{q}_i \cdot d_i}{q_i^{1-\alpha_i}} = \tilde{q}_i^{\alpha_i} q_i^{\alpha_i-1}, \quad S_i = + \int \tilde{q}_i^{\alpha_i} q_i^{\alpha_i-1} dq_i = \frac{1}{\alpha_i} \tilde{q}_i^{\alpha_i} q_i^{\alpha_i} = \frac{1}{\alpha_i} (\tilde{q}_i q_i)^{\alpha_i}$$

$$\Rightarrow S = \sum_{i=1}^n \frac{1}{\alpha_i} (\tilde{q}_i q_i)^{\alpha_i}$$

N24.9

$$1) \Pi(r) = -\frac{U_1}{r} + \frac{U_2}{r^2} \quad \text{в } \Psi_1(r, \theta, \varphi)$$

$$T = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\varphi}^2), \quad \varphi = \text{const} \Rightarrow$$

$$T = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2)$$

$$L = T - \Pi = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) + \frac{U_1}{r} - \frac{U_2}{r^2}$$

$$p_r = \frac{\partial L}{\partial \dot{r}} = m \dot{r} \quad \left\{ \begin{array}{l} \dot{r} = \frac{p_r}{m} \\ \dot{\theta} = \frac{p_\theta}{m r^2} \end{array} \right.$$

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta}$$

$$H = p q - L = \left(\frac{p_r}{m}; \frac{p_\theta}{m r^2} \right) \left(\frac{p_r}{m} \right) - \frac{m}{2} \left(\frac{p_r^2}{m^2} + r^2 \frac{p_\theta^2}{m^2 r^4} \right) - \frac{U_1}{r} + \frac{U_2}{r^2} = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2m r^2} - \frac{U_1}{r} + \frac{U_2}{r^2}$$

$$H(r, \theta, \frac{\partial S}{\partial r}, \frac{\partial S}{\partial \theta}) + \frac{\partial S}{\partial t} = 0 \quad - \text{уравн. Г-Я}$$

$$2) \Pi_4: \quad \alpha, \beta, \theta - \text{указ. координ.} \quad \theta = \alpha$$

$$p_\alpha = -\frac{\partial H}{\partial \alpha} = 0 \Rightarrow p_\alpha^2 = \text{const} = \alpha_1 \quad - \Pi_4$$

$$H = h = \text{const}; \quad S = S_0(\alpha, t) + S_1(\alpha, \alpha_1) + S_2(\alpha, \alpha_2)$$

$$0 = h + \frac{\partial S_0}{\partial t} \Rightarrow S_0 = -ht$$

$$\alpha_2 = \frac{\partial S}{\partial \alpha} = p_\alpha = \sqrt{\alpha_1} \Rightarrow S_1 = \sqrt{\alpha_1} \alpha$$

$$\frac{\partial S_2}{\partial r} = p_r = \left(h + \frac{U_1}{r} - \frac{U_2}{r^2} - \frac{\alpha_1}{2mr^2} \right) 2m$$

$$S_2 = \int \left(h + \frac{U_1}{r} - \frac{U_2}{r^2} - \frac{\alpha_1}{2mr^2} \right) 2m dr$$

$$S = -ht + \int 2m \left(h + \frac{U_1}{r} - \frac{U_2}{r^2} - \frac{\alpha_1}{2mr^2} \right) dr + \sqrt{\alpha_1} \alpha$$

$$3) \beta = \frac{\partial S}{\partial \alpha_1} = \beta(r, \alpha, \alpha_1) \quad \text{где } \beta \text{ невыраж. } r(\alpha, \beta, \alpha_1)$$

N24.43

$$H = \frac{p_1^2}{q_1^2} + \frac{1}{q_1^2} \left(\frac{p_2^2}{q_2^2} + \frac{p_3^2}{q_2^2 q_3^2} \right)$$

$$0 = H + \frac{\partial S}{\partial t} \quad - \Gamma - \Psi$$

$$\Pi_4: \quad H - \text{заменила} \quad \frac{p_3^2}{q_3^2} = \alpha_3, \quad \frac{p_2^2}{q_2^2} + \frac{\alpha_3^2}{q_2^2} = \frac{p_2^2 + \alpha_3^2}{q_2^2} = \alpha_2$$

$$\frac{p_1^2}{q_1^2} + \frac{\alpha_2}{q_1^2} = \frac{p_1^2 + \alpha_2}{q_1^2} = \alpha_1$$

$$S = S_0(\alpha, t) + S_1(q_1, \alpha_1) + S_2(q_2, \alpha_2) + S_3(q_3, \alpha_3)$$

Starten bei $\frac{\partial S_0}{\partial t} + H = 0$, $H = \alpha_1 \Rightarrow S_0 = -\alpha_1 t$

$$\frac{\partial S_1}{\partial q_1} = p_1 = \sqrt{q_1^2 \alpha_1 - \alpha_2}, \quad \frac{\partial S_2}{\partial q_2} = p_2 = \sqrt{\alpha_2 q_2^2 - \alpha_3^2}$$

$$\frac{\partial S_3}{\partial q_3} = p_3 = \alpha_3 q_3$$

$$S_1 = \int \sqrt{q_1^2 \alpha_1 - \alpha_2} dq_1, \quad S_2 = \int \sqrt{\alpha_2 q_2^2 - \alpha_3^2} dq_2, \quad S_3 = \int \alpha_3 q_3 dq_3 = \frac{\alpha_3 q_3^2}{2}$$

$$S = -\alpha_1 t + \int \sqrt{q_1^2 \alpha_1 - \alpha_2} dq_1 + \int \sqrt{\alpha_2 q_2^2 - \alpha_3^2} dq_2 + \int \alpha_3 q_3 dq_3$$

$S_0 + S_1$ $\beta_1 = \frac{\partial S}{\partial \alpha_1} = -t + \int \frac{\partial}{\partial \alpha_1} \sqrt{q_1^2 \alpha_1 - \alpha_2} dq_1 = -t + \int \frac{q_1^2}{2 \sqrt{q_1^2 \alpha_1 - \alpha_2}} dq_1 \Rightarrow$

$$2\beta_1 + 2t = \int \frac{q_1^2}{\sqrt{q_1^2 \alpha_1 - \alpha_2}} dq_1 \Rightarrow q_1(\alpha_1, \alpha_2, \beta_1, t)$$

$S_1 + S_2$ $\beta_2 = \frac{\partial S}{\partial \alpha_2} = \int \frac{\partial}{\partial \alpha_2} \sqrt{\alpha_2 q_2^2 - \alpha_3^2} dq_2 = \int \frac{q_2^2}{2 \sqrt{\alpha_2 q_2^2 - \alpha_3^2}} dq_2 =$

$$\beta_3 = \frac{\partial S}{\partial \alpha_3} = \frac{\partial (S_2 + S_3)}{\partial \alpha_3} = \int \frac{q_3^2}{2} - \int \frac{\alpha_3}{\sqrt{\alpha_2 q_2^2 - \alpha_3^2}} dq_2$$

N 24.66

$$H = H[\psi_1(q_1, p_1), \dots, \psi_n(q_n, p_n), t]$$

Mit: allgemeine n-paare

$$\psi_i(q_i, p_i) = \alpha_i$$

$$S = S_0(\alpha, t) + \sum_{i=1}^n S_i(q_i, \alpha_i)$$

$$\frac{\partial S_0}{\partial t} = -H \Rightarrow S_0 = -\int H dt$$

$$\frac{\partial S_i}{\partial q_i} = p_i = \psi_i(q_i, \alpha_i) \Rightarrow S_i = \int p_i(q_i, \alpha_i) dq_i$$

$$S = -\int H dt + \sum_{i=1}^n \int \psi_i(q_i, \alpha_i) dq_i$$

N 24.87

$$S = -\alpha_1 \int f(t) dt + \int \alpha_n f_n(q_n) dq_n + \sum_{i=1}^{n-1} \int (\alpha_i - \alpha_{i+1}) f_i(q_i) dq_i$$

$$H = -\frac{\partial S_0}{\partial t} = \alpha_1 \frac{\partial}{\partial t} \int f(t) dt = \alpha_1 f(t)$$

$$\frac{\partial S_i}{\partial q_i} = p_i = \alpha_i f_i(q_i) (\alpha_i - \alpha_{i+1}), \quad \alpha_i - \alpha_{i+1} = \frac{p_i^2}{f_i(q_i)}$$

$$\alpha_n = \frac{p_n^2}{f_n(q_n)}, \quad \alpha_1 = (\alpha_1 - \alpha_2) + (\alpha_2 - \alpha_3) + (\alpha_3 - \dots) + \alpha_n =$$

$$= \frac{p_1^2}{f_1(q_1)} + \frac{p_2^2}{f_2(q_2)} + \dots + \frac{p_{n-1}^2}{f_{n-1}(q_{n-1})} + \dots = \sum_{i=1}^{n-1} \frac{p_i^2}{f_i(q_i)} + \frac{p_n^2}{f_n(q_n)} =$$

$$= \sum_{i=1}^n \frac{p_i^2}{f_i(q_i)} \Rightarrow H = f(t) \sum_{i=1}^n \frac{p_i^2}{f_i(q_i)}$$

Ombem: $H = f(t) \sum_{i=1}^n \frac{p_i^2}{f_i(q_i)}$